Diamond DA40 Diamond Star (DA40)

Lycoming IO-360-M1A, 180 hp · 4 seats · Typical DA40 / DA40 XL / XLS (IO-360-M1A, constant-speed prop, G1000) AFM procedures; ESS BUS switch and long-range fuel transfer pump only where fitted.

V-speeds & limits

Vr	**59 KIAS (flaps T/O)**
Vx	**66 KIAS**
Vy	**73 KIAS**
Best glide	**88 KIAS (flaps UP, prop windmilling)**
Vfe	**108 KIAS (T/O) / 91 KIAS (LDG)**
Va / Vo	**108 KIAS (2,535 lb)**
Vno	**129 KIAS**
Vne	**178 KIAS**
Vs / Vso	**52 / 49 KIAS**
Vref	**70–73 KIAS (flaps LDG)**

Normal procedures

Before Start

Preflight inspection	**COMPLETE (fuel drains, oil 6–8 qt, tow bar removed)**
Passenger briefing	**COMPLETE (belts, canopy / rear door operation, exits)**
Seats, belts, harnesses	**ADJUST & LOCK**
Rudder pedals	**ADJUST & LOCK**
Canopy	**CLOSED & LOCKED (white marks aligned) or cooling gap position for taxi**
Rear door	**CLOSED & LOCKED**
Parking brake	**SET**
Fuel selector	**LEFT or RIGHT (fuller tank; never OFF)**
Throttle	**IDLE**
RPM lever	**HIGH RPM (full forward)**
Mixture	**IDLE CUT-OFF**
Alternate air	**CLOSED (pushed in)**
Avionics master	**OFF**
Electric master (BAT)	**ON**
ALT switch	**ON**
ESS BUS switch (if fitted)	**OFF**
Annunciators / G1000	**CHECK (no red, DOORS message off)**
Fuel quantity	**CHECK (vs visual check)**
Flaps	**CHECK operation, set UP**
Strobes / position lights	**ON**

Engine Start

Propeller area	**CLEAR**
Throttle	**OPEN approx 1/2 in**
Electric fuel pump	**ON**
Mixture	**RICH 3–5 s (prime, cold engine), then IDLE CUT-OFF**
Starter	**ENGAGE (max 10 s, then 20 s cooling)**
Mixture	**RICH when engine fires**
Throttle	**1,000–1,200 RPM**
Oil pressure	**CHECK rising within 30 s (green 55–95 psi)**
Electric fuel pump	**OFF (fuel pressure steady)**
Alternator	**CHECK charging (ammeter positive, no LOW VOLTS)**
Avionics master	**ON**
Engine instruments	**CHECK**
Pitot heat	**TEST then OFF**

Before Taxi

Flight instruments / G1000	**CHECK & SET (altimeter, HSI, CDI)**
Radios / transponder	**SET**
Trim	**SET for takeoff (neutral mark)**
Flaps	**T/O**
Parking brake	**RELEASE**
Brakes	**CHECK immediately after moving**

Nosewheel is free-castoring: steer with differential braking at low speed, rudder becomes effective above a brisk walking pace. Avoid riding the brakes – brake heat damages the composite wheel fairings.

Turn coordinator / HSI	**CHECK during turns**

Before Takeoff

Parking brake	**SET (into wind, clear area behind)**
Flight controls	**FREE & CORRECT**
Fuel selector	**FULLER TANK (check LEFT or RIGHT)**
Mixture	**RICH (lean as required above 5,000 ft)**
Throttle	**2,000 RPM**
Magnetos	**CHECK L / R (max drop 175 RPM, max diff 50 RPM)**
RPM lever	**CYCLE to LOW RPM – RPM drops, oil pressure stable – then HIGH RPM**
Alternate air	**OPEN – RPM drop – CLOSE**
Engine instruments	**CHECK green (oil temp min 75°F for takeoff)**
Ammeter / voltage	**CHECK charging**
Throttle	**IDLE check (600–800 RPM), then 1,000 RPM**
Electric fuel pump	**ON**
Flaps	**T/O (check indication)**
Trim	**SET takeoff**
Canopy	**CLOSED & LOCKED (no gap, both handles)**
Rear door	**LOCKED (DOORS annunciation off)**
Pitot heat / lights	**AS REQUIRED**
Transponder	**ALT**
Takeoff briefing	**COMPLETE (Vr 59, Vy 73, abort / EFATO plan)**

After Takeoff / Climb

Airspeed	**Vy 73 KIAS (obstacles: Vx 66 KIAS)**
Flaps	**UP (above 71 KIAS, positive rate)**
Electric fuel pump	**OFF above 1,000 ft AGL**
Power	**FULL THROTTLE / 2,400–2,700 RPM (cruise climb 2,400 RPM, 90 KIAS)**
Mixture	**RICH (lean above 5,000 ft)**
Engine instruments	**CHECK (CHT, oil temp max 245°F)**

Cruise

Power	**SET (typ. 2,200–2,400 RPM, max 75% continuous)**
Mixture	**LEAN (peak EGT then enrich 50–100°F, or per AFM table)**
Fuel selector	**SWITCH tanks every 30–60 min, keep imbalance low**
Fuel transfer pump (long-range tanks)	**ON when main tank low, OFF when transfer complete**
Fuel quantity / endurance	**CHECK vs flight plan**
Engine instruments	**CHECK**

Descent

Altimeter	**SET**
Mixture	**ENRICH as altitude decreases**
Power	**REDUCE gradually (avoid shock cooling)**
Fuel selector	**FULLER TANK**
Approach briefing	**COMPLETE (ATIS, approach, speeds)**

Before Landing

Seats / belts	**SECURE**
Fuel selector	**FULLER TANK**
Electric fuel pump	**ON**
Mixture	**RICH**
RPM lever	**HIGH RPM**
Flaps	**T/O below 108 KIAS, LDG below 91 KIAS**
Airspeed	**FINAL 70–73 KIAS (flaps LDG); 76 KIAS flaps T/O**
Landing light	**ON**

After Landing

Flaps	**UP**
Electric fuel pump	**OFF**
Pitot heat	**OFF**
Transponder	**STBY / as required**
Lights	**AS REQUIRED**
Canopy	**COOLING GAP position permitted for taxi**

Shutdown

Parking brake	**SET**
Throttle	**1,000–1,200 RPM, 1 min cool-down**
Avionics master	**OFF**
Mixture	**IDLE CUT-OFF**
Magnetos	**OFF (after prop stops), key removed**
ALT / Electric master	**OFF**
Fuel selector	**LEFT or RIGHT (leave on a tank)**
Canopy	**CLOSED, sun cover installed**
Pitot cover / chocks / tie-downs	**INSTALL**

Abnormal procedures

Alternator Failure / LOW VOLTS

LOW VOLTS or ALT FAIL caution, ammeter discharge

ALT circuit breaker ..... **CHECK IN**

ALT switch ..... **OFF, then ON (reset once)**

▶ Alternator restored?

Yes:

Electrical load ..... **MONITOR**

No:

ALT switch ..... **OFF**

ESS BUS switch (if fitted) ..... **ON (powers PFD, COM1/NAV1, AHRS, ADC, transponder)**

Non-essential loads ..... **OFF (lights, pitot heat if not needed, MFD if not on ESS BUS)**

Battery ..... **MONITOR voltage (approx 30 min at reduced load – verify per AFM)**

Deferred: Alternator failed – on battery, limited endurance. Use flaps once only on final; land as soon as practical. ..... **NOTED**

Low Oil Pressure / High Oil Temperature

Oil pressure below 55 psi, oil temperature above 245°F or CHT above 500°F

Oil pressure / temperature / CHT ..... **CHECK all, cross-check**

▶ Which indication?

Low pressure:

▶ Oil temperature rising or pressure below 25 psi?

Yes:

Power ..... **REDUCE to minimum required**

Landing site ..... **SELECT, expect engine failure**

Deferred: Low oil pressure – land as soon as possible, engine failure possible. ..... **NOTED**

No – temperature normal:

Flight ..... **CONTINUE, monitor closely**

Deferred: Suspect oil pressure indication – land as soon as practical. ..... **NOTED**

High temperature:

Mixture ..... **ENRICH**

Airspeed ..... **INCREASE (lower nose)**

Power ..... **REDUCE**

▶ Temperature returning to green?

Yes:

Flight ..... **CONTINUE, monitor**

No:

Power ..... **MINIMUM practical**

Deferred: Engine overheating – land as soon as practical, expect possible power loss. ..... **NOTED**

Rough Running / Loss of Power

Mixture ..... **ADJUST (richer, then leaner)**

Electric fuel pump ..... **ON**

Fuel selector ..... **SWITCH tanks**

Alternate air ..... **OPEN (induction icing or blockage)**

Magnetos ..... **CHECK L / R individually, BOTH**

▶ Engine running smoothly?

Yes:

Configuration ..... **KEEP the working combination**

Deferred: Engine ran rough – land as soon as practical, maintenance check. ..... **NOTED**

No:

Power ..... **REDUCE to smoothest setting**

Land ..... **AS SOON AS POSSIBLE (prepare for engine failure)**

Low Fuel Pressure / Fuel Flow Fluctuation

Fuel pressure below 14 psi or FUEL PRESS caution

Electric fuel pump ..... **ON**

Fuel selector ..... **SWITCH to fuller tank**

Mixture ..... **RICH**

Fuel quantity ..... **CHECK both tanks**

▶ Fuel pressure restored?

Yes:

Electric fuel pump ..... **LEAVE ON**

Deferred: Electric fuel pump on continuously – engine-driven pump suspect. ..... **NOTED**

No:

Fuel transfer pump (if fitted) ..... **ON**

Land ..... **AS SOON AS POSSIBLE, expect engine failure**

Flap Failure

FLAPS circuit breaker ..... **CHECK / RESET once**

Flap switch ..... **CYCLE**

▶ Flaps operable?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Approach ..... **FLAPS UP 78–80 KIAS / T/O 76 KIAS, flat approach**

Deferred: Flaps inoperative – flaps-up landing, increased approach speed and landing distance. ..... **NOTED**

Canopy or Rear Door Open in Flight

DOORS caution or canopy / door unlatched

Airspeed ..... **REDUCE below 108 KIAS (Vfe T/O)**

Canopy / door ..... **DO NOT attempt to close in flight**

The canopy will stay in the first (cooling gap) position; the rear door may open a few inches. Aircraft remains controllable, expect noise and buffet.

Loose objects / belts ..... **SECURE**

Landing ..... **NORMAL, flaps T/O or LDG**

Deferred: Canopy / door open – land as soon as practical, keep speed below 108 KIAS. ..... **NOTED**

PFD / AHRS / ADC Failure

Display backup (red button) ..... **PRESS – reversionary mode on MFD**

Standby instruments ..... **USE (airspeed, attitude, altimeter, compass)**

Autopilot ..... **OFF if AHRS failed**

PFD / AHRS circuit breakers ..... **CHECK**

▶ Flight in IMC?

Yes:

ATC ..... **ADVISE, request no-gyro assistance / nearest VMC**

Deferred: Primary flight display failed – partial-panel, request radar vectors, avoid IMC. ..... **NOTED**

No:

Deferred: PFD / AHRS failed – standby instruments only. ..... **NOTED**

Propeller Overspeed

RPM above 2,700 and not responding to RPM lever

Throttle ..... **REDUCE**

RPM lever ..... **LOW RPM, then set 2,700 or below**

Airspeed ..... **REDUCE (raise nose)**

Oil pressure ..... **CHECK (governor needs oil pressure)**

▶ RPM controllable?

Yes:

Power ..... **SET to keep RPM in limits**

Deferred: Propeller governor fault – land as soon as practical. ..... **NOTED**

No:

Power / airspeed ..... **USE to hold RPM below 2,700**

Land ..... **AS SOON AS POSSIBLE**

Static Source Blocked / Erroneous Airspeed

Pitot heat ..... **ON**

Alternate static source ..... **OPEN (if fitted) / crack cabin vent**

G1000 ADC ..... **CROSS-CHECK with GPS ground speed and standby altimeter**

Attitude / power ..... **FLY known pitch and power settings**

Deferred: Pitot-static unreliable – fly attitude and power, add margin to approach speed. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

Throttle (memory) ..... **IDLE**

Brakes (memory) ..... **APPLY (differential for directional control)**

Flaps ..... **UP**

Mixture ..... **IDLE CUT-OFF**

Fuel selector ..... **OFF**

Magnetos ..... **OFF**

Electric master ..... **OFF**

Engine Failure after Takeoff

Airborne, insufficient runway remaining

LAND IMMEDIATELY

- Airspeed (memory) ..... **73 KIAS (flaps UP) / 70 KIAS (flaps T/O)**
- Fuel selector (memory) ..... **SWITCH tanks**
- Electric fuel pump (memory) ..... **ON**
- Mixture (memory) ..... **RICH**
- Alternate air (memory) ..... **OPEN**

Below approx 500 ft AGL land straight ahead with only small heading changes. Do not attempt a turn back.

- Flaps ..... **LDG when landing assured**
- Mixture ..... **IDLE CUT-OFF**
- Fuel selector ..... **OFF**
- Magnetos ..... **OFF**
- Electric master ..... **OFF**
- Canopy ..... **UNLATCH (cooling gap) before touchdown**

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

- Airspeed (memory) ..... **88 KIAS (best glide, flaps UP)**
- Fuel selector (memory) ..... **SWITCH tanks**
- Electric fuel pump (memory) ..... **ON**
- Mixture (memory) ..... **RICH**
- Alternate air (memory) ..... **OPEN**
- Landing site ..... **SELECT, turn towards it**

▶ Restart attempt practicable?

Yes – restart:

- Magnetos ..... **BOTH (check)**
- Throttle ..... **1/2 in open**
- Mixture ..... **RICH**
- Starter ..... **ENGAGE if propeller has stopped**
- RPM lever ..... **HIGH RPM once running**

▶ Engine restarted?

Yes:

- Power ..... **AS REQUIRED (minimum needed)**
- Engine instruments ..... **MONITOR**
- Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel or induction problem. ..... **NOTED**

No:

Continue with forced landing.

No – forced landing:

Forced landing without power:

- Mayday / 7700 ..... **TRANSMIT**
- Seats / belts ..... **SECURE**
- Flaps ..... **LDG when landing assured**
- Airspeed ..... **70–73 KIAS on final**
- Mixture ..... **IDLE CUT-OFF**
- Fuel selector ..... **OFF**
- Magnetos ..... **OFF**
- Electric master ..... **OFF (when no longer required)**
- Canopy ..... **UNLATCH before touchdown**

Engine Fire in Flight

LAND IMMEDIATELY

Fuel selector (memory)	**OFF**
Mixture (memory)	**IDLE CUT-OFF**
Electric fuel pump (memory)	**OFF**
Throttle (memory)	**FULL OPEN (burn remaining fuel in lines)**
Cabin heat / defrost (memory)	**OFF**
Airspeed	**INCREASE (sideslip to keep flames from cabin)**
Magnetos	**OFF when engine stops**
Forced landing	**EXECUTE (do not attempt restart)**

Engine Fire During Start

Starter (memory)	**CONTINUE cranking (draw fire into engine)**
Mixture (memory)	**IDLE CUT-OFF**
Throttle (memory)	**FULL OPEN**
Fuel selector	**OFF**
Electric fuel pump	**OFF**
Magnetos / Electric master	**OFF**
Fire extinguisher	**OBTAIN**
Aircraft	**EVACUATE**

Cabin / Electrical Fire

LAND AS SOON AS POSSIBLE

Electric master (memory)	**OFF**
Avionics master (memory)	**OFF**
Cabin heat / fresh air	**CLOSE**
Fire extinguisher	**ACTIVATE**

After discharge, open fresh air vents / canopy cooling gap only once fire is out.

▶ Fire out and electrical power required?

Yes:

Circuit breakers	**CHECK – do not reset tripped breakers**
ESS BUS (if fitted)	**ON (essential bus only)**
Electric master	**ON**
Avionics	**ON one at a time, check for smoke**
Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable.	**NOTED**

No:

Continue with electrics off. Standby instruments and magnetic compass only; no flaps, no trim indication.

Deferred: All electrics off – flaps and electric trim unavailable, flaps-up landing (approx 80 KIAS), no radios.	**NOTED**

Emergency Descent

Throttle (memory)	**IDLE**
RPM lever (memory)	**HIGH RPM**
Flaps (memory)	**UP**
Airspeed (memory)	**178 KIAS smooth air / 129 KIAS turbulence**
Mixture	**ENRICH during descent**

Inadvertent Icing Encounter

LAND AS SOON AS PRACTICAL

Pitot heat (memory)	**ON**
Alternate air (memory)	**OPEN**
Exit icing conditions	**TURN BACK / CHANGE ALTITUDE**
Power	**INCREASE to prevent ice on propeller**
Cabin heat / defrost	**FULL**
Autopilot	**OFF (monitor trim and control feel)**
Airspeed	**KEEP HIGH, avoid steep turns**
Landing	**FLAPS UP or T/O, approach 80–90 KIAS**
Deferred: Residual airframe ice – flaps-up / T/O landing, increased approach speed, longer runway.	**NOTED**

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist 38b19394 · 2026-11-07